# Geometry of anti-Poisson involutions in the deformations and resolutions of Kleinian singularities

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Yale/JHU

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7 - Lie Theory and Generalizations

**Overview:**

$$\begin{array}{ccc} \text{minimal} & & \text{Kleinian} \\ \text{resolution} & & \text{singularity} \\ \widetilde{X} & \xrightarrow{\pi} & X \circlearrowleft \theta \end{array}$$

anti-Poisson
involution

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minimal resolution | Kleinian singularity

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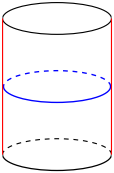
**Extended Goal:** Replace $X$ by its Poisson deformation $X_\lambda,\ \lambda \in \mathfrak{h}/W$, and describe families of Lagrangian subvarieties $X^\theta_\lambda$ and $\pi^{-1}(X^\theta_\lambda)$.

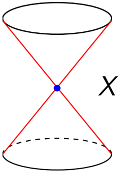
# An example

Type $A_1$ Kleinian singularity: $X = \operatorname{Spec} \mathbb{C}[x, y, z]/(xy - z^2)$
Anti-Poisson involution $\theta : x \leftrightarrow y$

$\widetilde{X} = T^*\mathbb{P}^1$



$\xrightarrow{\pi}$



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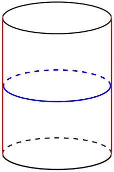
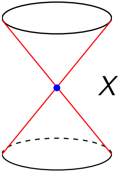
$\widetilde{X} = T^*\mathbb{P}^1$  $\xrightarrow{\pi}$  $X = \mathcal{N}$, nilpotent cone of $\mathfrak{sl}_2(\mathbb{C})$

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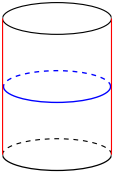

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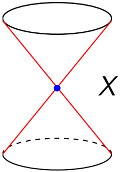

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Poisson deformation, $X_\lambda = \operatorname{Spec}\mathbb{C}[x,y,z]/(xy - z^2 - \lambda) \curvearrowleft \theta$
$X_\lambda^\theta = \operatorname{Spec}\mathbb{C}[x,z]/(x^2 - z^2 - \lambda) \simeq \mathbb{C}^*$ if $\lambda \neq 0$.

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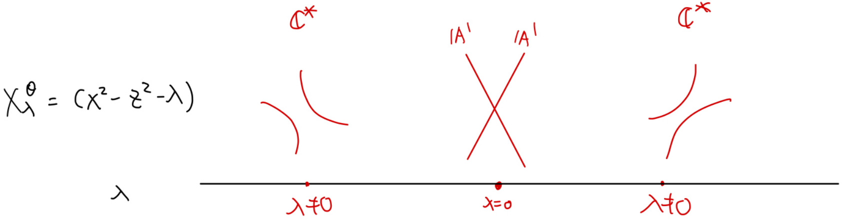
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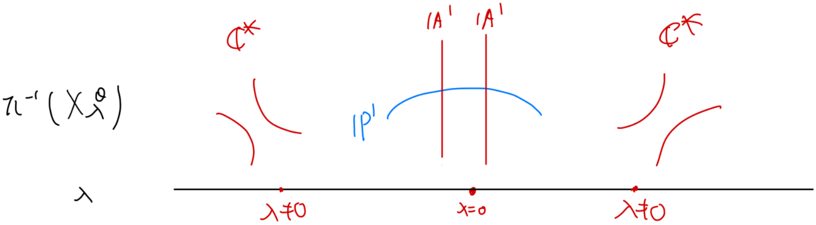
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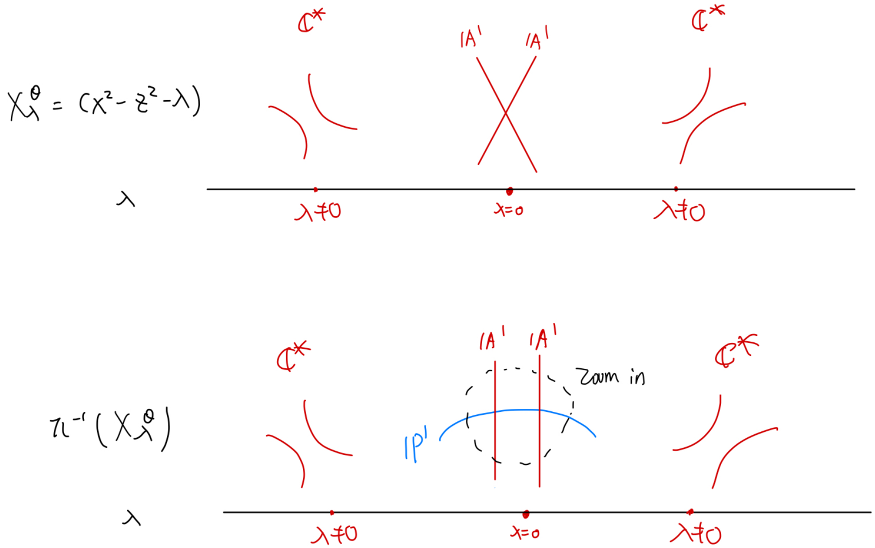

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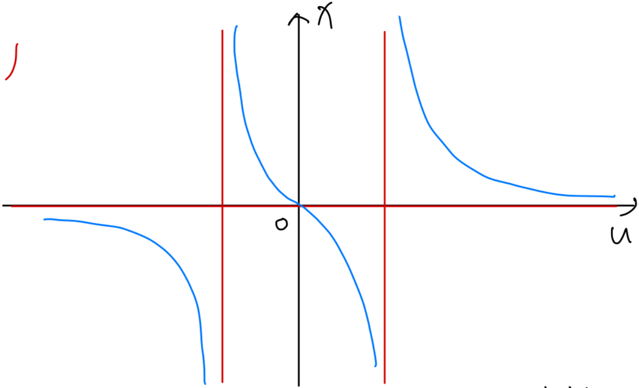


$$x(u^2-1) = 2\lambda u \Rightarrow x = \frac{2\lambda u}{u^2-1}$$

$$z = xu - \lambda$$

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## Example

When $\Gamma = \{\pm I_2\}$, we have $\mathbb{C}[u, v]^\Gamma =$ even degree polynomials $=$ $\mathbb{C}[x = u^2, y = v^2, z = uv] = \mathbb{C}[x, y, z]/(xy - z^2)$.

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## Fact (Klein, 1884)

$\mathbb{C}^2/\Gamma \hookrightarrow \mathbb{C}^3$ (one relation), and $\mathbb{C}^2/\Gamma$ has an isolated singularity at 0.

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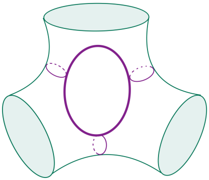

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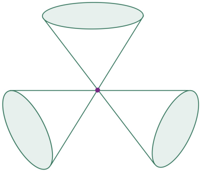

$D_4$ singularity
$x^3 + xy^2 + z^2 = 0$

# Anti-Poisson involutions & fixed point loci

Set $X := \mathbb{C}^2/\Gamma$. The algebra of functions $\mathbb{C}[X] = \mathbb{C}[u, v]^\Gamma$ is a graded Poisson algebra with **Poisson bracket**

$$\{f_1, f_2\} = \frac{\partial f_1}{\partial u}\frac{\partial f_2}{\partial v} - \frac{\partial f_1}{\partial v}\frac{\partial f_2}{\partial u}.$$

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**Definition:** An **anti-Poisson involution** of $X = \mathbb{C}^2/\Gamma$ is a graded algebra involution $\theta\colon \mathbb{C}[X] \to \mathbb{C}[X]$ such that

$$\theta(\{f_1, f_2\}) = -\{\theta(f_1), \theta(f_2)\}, \ \forall\ f_1, f_2 \in \mathbb{C}[X].$$

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**Definition:** The **fixed point locus** is $X^\theta := \operatorname{Spec} \mathbb{C}[X]/I$, where $I = (\theta(f) - f,\ f \in \mathbb{C}[X])$.

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## Proposition 1 (H.)

- There are finitely many anti-Poisson involutions on $\mathbb{C}^2/\Gamma$ up to conjugation by graded Poisson automorphisms.
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**Example:** Type $A_n$ singularity $X = \operatorname{Spec} \mathbb{C}[x, y, z]/(xy - z^{n+1})$. Then $\theta$ swapping $x \leftrightarrow y$ is an anti-Poisson involution. We have

$X^\theta = \operatorname{Spec} \mathbb{C}[x, y, z]/(xy - z^{n+1}, x - y) \simeq \operatorname{Spec} \mathbb{C}[x, z]/(x^2 - z^{n+1})$,

which is a union of two $\mathbb{A}^1$'s when $n$ is odd, a cusp when $n$ is even.

# Examples in type A

**Example:** Anti-Poisson involutions for type $A_n$ Kleinian singularities.

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|---|---|---|---|
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| involution of $\mathfrak{sl}_{n+1}$ | $M \mapsto -M^t$ | $\mathrm{Ad}(I_{\lceil \frac{n+1}{2} \rceil, \lfloor \frac{n+1}{2} \rfloor})$ | $\mathrm{Ad}(I_{\frac{n+3}{2}, \frac{n-1}{2}})$ |

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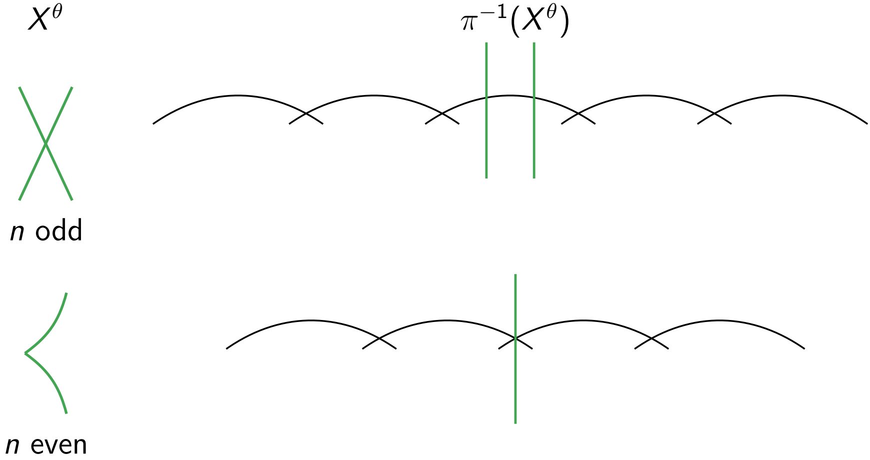
# Preimage of fixed point loci under minimal resolutions

Recall $\pi \colon \widetilde{X} \to X$ minimal resolution. $0 \in X^\theta \Rightarrow \pi^{-1}(0) \subset \pi^{-1}(X^\theta)$.

**Example:** Consider type $A_n$ singularities with $\theta$ swapping $x \leftrightarrow y$.

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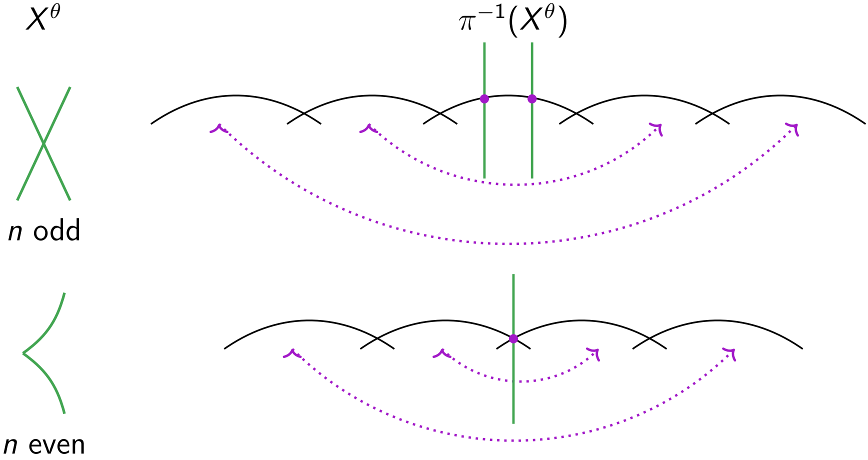


Lift the action of $\theta$ to the $\mathbb{P}^1$'s

# Lift anti-Poisson involutions

$\pi\colon \widetilde{X} \to X$, minimal resolution, and $\theta$ an anti-Poisson involution of $X$.

**Theorem (H., 2025)**

There exists a unique anti-symplectic involution $\tilde{\theta} : \widetilde{X} \to \widetilde{X}$ such that $\pi \circ \tilde{\theta} = \theta \circ \pi$. It can be constructed via involutions of quiver varieties.

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Write $\pi^{-1}(X^\theta) = \sum_{j=1}^{m} 1 \cdot L_j + \sum_{i=1}^{n} a_i C_i$ as a divisor, with $L_j \simeq \mathbb{A}^1$, $C_i \simeq \mathbb{P}^1$. There are methods to determine the multiplicities $a_i$.

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Let $\mathfrak{g}$ be a simple Lie algebra of types ADE and $\mathcal{N}$ its nilpotent cone. Take $e \in \mathbb{O}_{\mathrm{subreg}}$ and consider the Slodowy slice $S = e + \mathfrak{z}_{\mathfrak{g}}(f)$.

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**Question:** Can we extend the action of $\theta$ to $X_\lambda$, for $\lambda \neq 0$?

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**Fact:** The Slodowy slice $S$ is the universal graded Poisson deformation of Kleinian singularity $S \cap \mathcal{N} \simeq X$.

Consider the morphism $\varphi\colon S \to \mathfrak{h}/W$ defined as the composition

$$\varphi\colon S \hookrightarrow \mathfrak{g} \to \mathfrak{g}/\!/G \simeq \mathfrak{h}/W.$$

Zero fiber: $\varphi^{-1}(0) = S \cap \mathcal{N} \simeq X$ is the Kleinian singularity.
Other fibers: $X_\lambda := \varphi^{-1}(\lambda)$ is a filtered Poisson deformation of $X$.

**Question:** Can we extend the action of $\theta$ to $X_\lambda$, for $\lambda \neq 0$?

## Proposition 2 (H.)

$\theta$ naturally acts on $S$ and $\mathfrak{h}/W$ by the universal property. The action of $\theta$ extends to $X_\lambda$ iff $\lambda \in (\mathfrak{h}/W)^\theta$.

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# Lagrangians in deformations of Kleinian singularities

**Question:** How to describe $X_\lambda^\theta$ and $\pi^{-1}(X_\lambda^\theta)$, for $\lambda \neq 0$?

- Step 1: Determine the singular points of $X_\lambda$. Each is a Kleinian singularity of smaller type.
- Step 2: Identify the local Type of the anti-Poisson involution near each singular point.
- Step 3: Describe the local intersection pattern of projective lines and affine curves, and then connect these local pieces to obtain a global picture.

# Examples

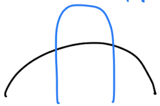
**Example:** An $A_2$ deformation with singularity type $A_1$

$$X_\lambda = \operatorname{Spec} \mathbb{C}[x, y, z] / (xy - (z-1)^2(z+2)) \qquad \theta : x \mapsto y$$

$$X_\lambda^\theta = (x^2 - (z-1)^2(z+2)) \qquad \pi^{-1}(X_\lambda^\theta)$$

nodal cubic

$\mathbb{A}^1$ (normalization of nodal cubic)



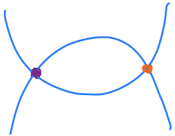
$\mathbb{P}^1$

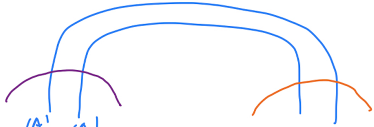
# Examples

**Example:** An $A_3$ deformation with singularity type $A_1 \times A_1$

$$X_\lambda = \mathrm{Spec}\, \mathbb{C}[x, y, z] / (xy - (z-1)^2(z+1)^2) \qquad \theta : x \leftrightarrow y$$

$$X_\lambda^\theta = (x^2 - (z-1)^2(z+1)^2) \qquad\qquad \pi^{-1}(X_\lambda^\theta)$$



Thank you!